

Worksheet — Units, Dimensions and Vectors (NIO Official)

Total marks: 30 · Questions: 10

Instructions: This is the NIOS Senior Secondary Physics (312) official Worksheet-1 for Lesson 1. Attempt all questions. Most questions are open-ended and inquiry-based — look around you, measure, and explain in your own words. Show your reasoning for numerical questions.

Q1. The scope of physics is very wide. It covers a vast variety of natural phenomena. Observe your surroundings and explain ONE application of physics in day-to-day life. [2 marks]

Q2. Observe your surroundings and mention any FIVE reasons why you need to measure. [2 marks]

Q3. Pick any FIVE objects from your surroundings with different shapes/dimensions. Make a table with columns: Name of the Object, Length/Breadth/Height, Weight/Area/Volume, Dimensional Formula. (Sample row: Matchbox, 5cm/3cm/2cm, 25 g / – / 30 cm³, L³) [3 marks]

Q4. In ancient times when there was no standard system of measurement, human body parts (foot, palm, span, cubit, digits) were used as units. Explain in your own words why human body parts were rejected as units for measurement. [3 marks]

Q5. For measurement, each physical quantity is assigned an SI unit (e.g. metre for distance). It can be converted in different units by using prefixes for powers of ten (e.g. 10^{-2} m = centimetre, 10^3 m = kilometre). Explain why we need different units for the same physical quantity. [2 marks]

Q6. Pick any object from home or school and measure it with a suitable scale. After measuring, convert the measured units into THREE different units of that quantity. [2 marks]

Q7. Give an example of: (a) a physical quantity which has neither unit nor dimensions; (b) a physical quantity which has a unit but no dimensions; (c) a constant which has no unit; (d) a constant which has a unit. [3 marks]

Q8. Mention 3 scalar quantities and 3 vector quantities from your surroundings. Write your comments on why scalar quantities are different from vector quantities, and also why we need vectors. [3 marks]

Q9. Push your study table in a certain direction with a force of magnitude 10 N. At the same time, ask someone to push the same table with a force of magnitude 15 N. Calculate the magnitude of the resultant force on the table and its direction when the forces are acting (i) in the same direction, and (ii) in opposite directions. [3 marks]

Q10. Find the area of the triangle whose vertices are $A(3, -1, 2)$, $B(1, -1, -3)$ and $C(4, -3, 1)$ using a suitable vector product. [7 marks]

Answer Key

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Q1. The scope of physics is very wide. It covers a vast variety of natural phenomena. Observe your surroundings and explain ONE application of physics in day-to-day life.

Answer: One example: the working of a pressure cooker. Water boils at a temperature that depends on the pressure above it (a thermodynamics result). Sealing the cooker raises the internal pressure as steam builds up, which raises the boiling point of water from 100 °C to around 120 °C — food cooks faster because chemical reactions accelerate with temperature. The safety valve uses Newton's laws (force balance between steam pressure and the weight on the valve) to release excess pressure. Other valid examples: lever in a pair of scissors; convex mirror at a road junction; electric kettle; bicycle gears.

Look for an example that names a real physical principle (heat, pressure, optics, mechanics, electricity, etc.) and explains how the device exploits it.

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Q2. Observe your surroundings and mention any FIVE reasons why you need to measure.

Answer: Five reasons to measure: (1) to compare two quantities objectively (e.g. is this room hotter than that one?); (2) to share a precise description with others without ambiguity (a recipe needs '200 g flour', not 'some flour'); (3) to check whether something meets a requirement (does this rod fit the slot?); (4) to track change over time (height of a child, weight of a person); (5) to design and build things that fit together (machine parts, buildings, electronic circuits). Measurement is the language in which physics is expressed.

Any five sensible reasons that explain why measurement is needed in daily life or in science.

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Q3. Pick any FIVE objects from your surroundings with different shapes/dimensions. Make a table with columns: Name of the Object, Length/Breadth/Height, Weight/Area/Volume, Dimensional Formula. (Sample row: Matchbox, 5cm/3cm/2cm, 25 g / – / 30 cm³, L³)

Answer: Sample table: (1) Matchbox: 5/3/2 cm, 25 g/–/30 cm³, dimensions of volume = L³. (2) Eraser: 4/2/1 cm, 8 g/–/8 cm³, L³. (3) Book: 25/18/2 cm, 350 g/450 cm²/900 cm³, L³ (volume) or L² (area). (4) Cricket ball: diameter 7 cm, 160 g/154 cm²/180 cm³ approx, L³. (5) Plate: diameter 25 cm, 200 g/491 cm²/–, L² (area). Each row should show the actual measurement and identify whether the derived quantity has dimensions of length L, area L², volume L³ or mass M.

Look for sensible measurements with correct units, and the correct dimensional formula (M for mass, L for length, L² for area, L³ for volume).

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Q4. In ancient times when there was no standard system of measurement, human body parts (foot, palm, span, cubit, digits) were used as units. Explain in your own words why human body parts were rejected as units for measurement.

Answer: Human body parts were rejected as units because they are not reproducible: (1) different people have different-sized hands, feet and arms — a 'foot' for one person is not the same as a 'foot' for another; (2) even for the same person, body parts grow with age and change with weight; (3) two measurements made by two people cannot be reliably compared; (4) trade and engineering need universally agreed standards that everyone can reproduce. The SI system was developed precisely to provide internationally agreed, reproducible standards (e.g. the metre defined by the speed of light) that are independent of any individual.

Key idea: a good unit must be reproducible and universal. Body parts vary across individuals and over time, making them unsuitable.

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Q5. For measurement, each physical quantity is assigned an SI unit (e.g. metre for distance). It can be converted in different units by using prefixes for powers of ten (e.g. 10^{-2} m = centimetre, 10^3 m = kilometre). Explain why we need different units for the same physical quantity.

Answer: We use different-sized units for the same quantity because the things we measure vary across many orders of magnitude. The diameter of an atom is about 10^{-10} m and the distance to a galaxy is about 10^{22} m — these cannot both be conveniently expressed in metres. Using prefixes (nano-, milli-, kilo-, mega-, etc.) makes numbers easy to read, write and compute with: '5 nm' is friendlier than '0.000000005 m' and '300 km' is friendlier than '300000 m'. The prefix system in SI gives a compact, consistent way to scale to whatever range a quantity covers.

The answer should mention the very wide range of magnitudes encountered in physics and the convenience of decimal-prefix scaling.

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Q6. Pick any object from home or school and measure it with a suitable scale. After measuring, convert the measured units into THREE different units of that quantity.

Answer: Example: the length of a notebook measured with a 30 cm ruler is 25 cm. Convert: (1) 25 cm = 0.25 m (SI base unit); (2) 25 cm = 250 mm; (3) 25 cm = 2.5×10^{-4} km = 0.00025 km. (Or in non-SI: 25 cm $\approx$ 9.84 inches.) The same length is expressed in different magnitudes by choosing different unit sizes — the underlying physical length is unchanged.

Look for a measured value with units, followed by three valid conversions using either prefixes (mm, cm, m, km) or different unit systems (cm vs inch).

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Q7. Give an example of: (a) a physical quantity which has neither unit nor dimensions; (b) a physical quantity which has a unit but no dimensions; (c) a constant which has no unit; (d) a constant which has a unit.

Answer: (a) Refractive index — a pure ratio of speeds, no unit and no dimensions. (Also: strain, relative density.) (b) Plane angle (radian) and solid angle (steradian) — they have a unit but are dimensionless (radian = arc length / radius, a length ratio). (c) The fine-structure constant $\alpha \approx 1/137$ — a dimensionless physical constant. (Also: π , Reynolds number, Avogadro's number is sometimes counted here.) (d) Acceleration due to gravity $g \approx 9.8 \text{ m/s}^2$ (m s^{-2}); speed of light $c = 3 \times 10^8 \text{ m/s}$; Planck's constant h with unit J·s.

Distinguish between (a) and (b): radians have a NAMED unit but no fundamental dimension, while ratios like refractive index have no unit at all.

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Q8. Mention 3 scalar quantities and 3 vector quantities from your surroundings. Write your comments on why scalar quantities are different from vector quantities, and also why we need vectors.

Answer: Three scalars: temperature of the room (28 °C); mass of a book (350 g); time taken to walk to school (15 minutes). Three vectors: displacement when you walk from home to school (1.5 km north-east); velocity of a moving car (50 km/h east); force you apply to push a door (20 N along the door's normal). Scalars are fully described by a number plus a unit, with no direction; they can be added or subtracted using ordinary arithmetic. Vectors are described by both magnitude AND direction, and they combine by special rules (triangle law / component-wise addition). We need vectors because many physical quantities — force, velocity, acceleration, displacement, momentum — change their effect depending on direction; ignoring the direction would give wrong results.

Look for valid examples and an explanation of the magnitude-only vs magnitude-and-direction distinction, and the need for direction-aware arithmetic.

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Q9. Push your study table in a certain direction with a force of magnitude 10 N. At the same time, ask someone to push the same table with a force of magnitude 15 N. Calculate the magnitude of the resultant force on the table and its direction when the forces are acting (i) in the same direction, and (ii) in opposite directions.

Answer: (i) Same direction: resultant magnitude = $10 + 15 = 25 \text{ N}$ in the common direction of the two pushes. (ii) Opposite directions: resultant magnitude = $|15 - 10| = 5 \text{ N}$ in the direction of the larger force (the 15 N push). The table is pushed by 5 N in the direction of whoever is pushing harder. (General case: if the forces are at angle θ , the magnitude is $R = \sqrt{F_1^2 + F_2^2 + 2 F_1 F_2 \cos \theta}$.)

Apply $R = \sqrt{F_1^2 + F_2^2 + 2 F_1 F_2 \cos \theta}$ with $\theta = 0$ for parallel and $\theta = 180^\circ$ for anti-parallel.

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Q10. Find the area of the triangle whose vertices are A(3, -1, 2), B(1, -1, -3) and C(4, -3, 1) using a suitable vector product.

Answer: The area of the triangle equals half the magnitude of the cross product of two side-vectors. Let $\vec{AB} = \vec{B} - \vec{A} = (1-3, -1-(-1), -3-2) = (-2, 0, -5)$. Let $\vec{AC} = \vec{C} - \vec{A} = (4-3, -3-(-1), 1-2) = (1, -2, -1)$. Cross product $\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 0 & -5 \\ 1 & -2 & -1 \end{vmatrix} = \hat{i} [(0)(-1) - (-5)(-2)] - \hat{j} [(-2)(-1) - (-5)(1)] + \hat{k} [(-2)(-2) - (0)(1)] = \hat{i} [0 - 10] - \hat{j} [2 - (-5)] + \hat{k} [4 - 0] = -10\hat{i} - 7\hat{j} + 4\hat{k}$. Magnitude: $|\vec{AB} \times \vec{AC}| = \sqrt{(100 + 49 + 16)} = \sqrt{165}$. Area = $(1/2) \times \sqrt{165} \approx (1/2) \times 12.845 \approx 6.42$ square units.

Area of a triangle from three position vertices = $(1/2) |\vec{AB} \times \vec{AC}|$. The cross product needs careful sign tracking.

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